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A Novel Intelligent Managed Pressure Drilling System Based on Multi-Device Collaborative Control and Its Application

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Abstract

With the rapid development of automation and intelligence technology in petroleum engineering field, automated drilling equipment is gradually being popularized and promoted, drilling engineering is gradually moving towards intelligence. The closed-loop pressure intelligent managed is one of the key components of intelligent drilling. Therefore, a novel intelligent managed pressure drilling (MPD) system based on the automated drilling rig equipment should be developed for solving the problem of drilling safety and efficiency in narrow density windows formation.

Firstly, a novel three-layer architecture integrated control and communication system for collaborative control both MPD and automated drilling rig equipment was established, a multi-source real-time data interface integrating all automated equipment operating status and control parameters was developed in the meantime, based on which the hardware and software barriers for collaborative control between MPD and automated drilling rig equipment was cleared. In order to improve the pressure control precision during the entire process of different drilling and completion conditions, a comprehensive pressure control method was established, in which the choke movement was adjusted based on the 7-segment S-type and T-type acceleration control algorithm used servo electric actuator. At the same time, the drilling rig lifting system and circulation system control were integrated, through which the problem of coordinated control between pressure control equipment and automated drilling rig was solved, and the pressure response speed is improved by one order of magnitude. Moreover, a series of system control and simulation software were developed, through which not only the pressure but also the circulation flow, top drive motion status can be intelligent control easily.

The novel intelligent control MPD system has been applied in more than 20 shale oil and gas wells of SINOPEC, the results show that we can make precisely pressure control based on the novel MPD system and the entire process pressure control error is less than 30psi. Based on the intelligent MPD system, the NPT have been reduced more than 60%.

This paper developed a novel intelligent MPD system, which can achieve the entire process pressure control of drilling and completion based on collaborative control MPD and automated drilling rig

equipment. Through the application, the results show that it can provide technical support to reduce operating risk and improve the intelligent level of MPD, the novel MPD system can improve the success rate of narrow window formation drilling and reduce the drilling cost.

Keywords: intelligent drilling, automated drilling, collaborative control, multi-device linkage effect, intelligent MPD

Introduction

With the rapid development of automation and intelligence technology in petroleum engineering field, automated drilling equipment is gradually being popularized and promoted, drilling engineering is gradually moving towards intelligence. The closed-loop pressure intelligent managed is one of the key components of intelligent drilling. Therefore, a novel intelligent managed pressure drilling (MPD) system based on the automated drilling rig equipment should be developed for solving the problem of drilling safety and efficiency in narrow density windows formation.

Following the resolution of core technologies, CNPC and Sinopec achieved industrial-scale deployment of hydraulically actuated managed-pressure drilling (MPD) systems by 2011 (Sun et al., 2015). However, the shift of exploration targets towards deep-ultra-deep formations, shale oil/gas reservoirs, and other unconventional formations, which has introduced increasingly severe geomechanical constraints. Such as narrow drilling mud weight windows, high-pressure/high-temperature (HPHT) conditions. These conditions expose fundamental limitations in hydraulic MPD actuation: (1) insufficient control resolution (± 30 psi tolerance remains unattainable), (2) sluggish dynamic response due to fluid compressibility delays, (3) inadequate pressure containment and sealing integrity above 70 MPa, and (4) deficient real-time interoperability with automated drilling rigs for closed-loop control (Smith et al., 2018; Li et al., 2020). These limitations directly contribute to increased non-productive time (NPT) and escalating operational expenditures. Consequently, the development of next-generation, electrically driven precision pressure control technology, millisecond-level dynamic response, full HPHT compatibility, and seamless rig-level integration is imperative to overcome current performance ceilings in complex drilling environments (Liu et al., 2021).

Specifically, there are several shortcomings of the hydraulic pressure control equipment in its application process. Firstly, inherent limitations of hydraulic control systems restrict their application in high-precision pressure regulation. When using electro-proportional valves to modulate a hydraulically driven valve actuator, the compressibility of hydraulic fluid and its temperature-dependent viscosity introduce positioning errors that typically remain within ± 1 mm, accompanied by significant lag (typical response > 500 ms). In the high-pressure operating range (e.g., when the choke is near closed), fluid compression is magnified, producing pressure oscillations of 10%~15 % of the set-point (Merritt, 1967) and preventing pressure control within ± 0.3 MPa (Zhou et al., 2018). Ambient temperature excursions ($-20 \sim 60$ °C) further amplify viscosity drift, capping repeatability at ± 0.5 mm (Naseradinmousavi et al., 2016) and failing to meet the stringent requirement of ≤ 0.1 MPa pressure fluctuation for complex shale-gas formations (Smith et al., 2020).

Dynamic regulation challenges in specialized processes expose the response shortcomings of hydraulic equipment. In MPC(Managed Pressure Cementing) or kick-mitigation, strong nonlinear pressure disturbances arise from the interaction of multiphase fluids inside the wellbore (Hannegan, 2011), requiring the target pressure curve to be updated 2~3 times per second (Santos et al., 2019). Existing hydraulic MPD systems, limited by valve spool inertia and hydraulic transmission delay, exhibit an average response time of 45s (API RP 92, 2020), far exceeding the ≤ 5 s real-time adjustment window demanded by the process (Chau et al., 2022). Consequently, early kick detection cannot be translated into timely suppression of annular pressure peaks (Godhavn et al., 2016).

Automation-integration bottlenecks impede the advancement of intelligent drilling. Modern electric rigs have achieved coordinated control among the driller's console, top-drive, and auto-driller (ISO 21882:2023), yet hydraulic pressure-management equipment still relies on stand-alone PLCs that provide only basic HMI monitoring of choke position and flow-meter readings (Saputelli et al., 2019). Although pressure feedback loops are implemented using PID tuning based on MPD-engineer experience, the absence of industrial protocols such as Modbus/TCP or OPC UA (IEC 61158) prevents real-time data exchange with the rig's intelligent decision system (e.g., automated well-control algorithms) (Booth et al., 2021), thereby hindering the field deployment of adaptive MPD (Gravdal et al., 2020).

To address the above challenges, breakthroughs were achieved in the R&D of core components, key methodologies, system integration, and process optimization for precision pressure-control equipment, culminating in the development of a new-generation electric MPD system that is "fast-responding, high-precision, intelligent, and full-process".

Three-layer integrated communication electro-hydraulic system

To enable integrated communication within the automatic MPD system, between MPD and the automated drilling rig, a multi-layer communication-fusion electro-hydraulic architecture was designed, as shown in the Figure 1. A proprietary three-layer electro-hydraulic architecture was developed to integrate automated drilling rigs with automatic managed-pressure drilling (MPD) systems, eliminating the hardware barriers to data exchange inside the MPD equipment modules, between the MPD and drilling-rig systems, and between the MPD and the intelligent drilling integrated control center.

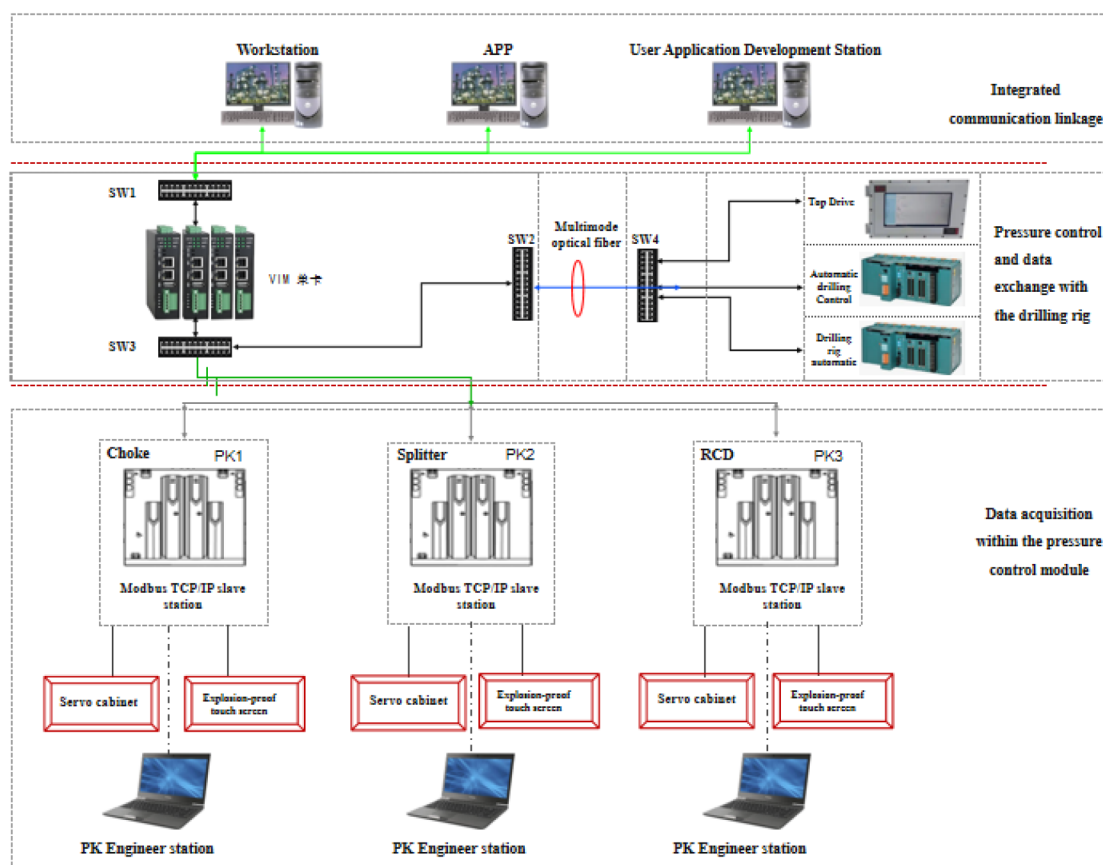


Figure 1—Multi-layer Integrated Control Electro-Hydraulic System

Design and matching of internal monitoring electro-hydraulic systems for automatic managed-pressure drilling

Dedicated electro-hydraulic systems were developed for the electric choke module and the influx/loss monitoring module, respectively. Each module is equipped with reserved interfaces to enable seamless data exchange between them.

Choke-module electro-hydraulic system : The system acquires and controls in real time the servo-choke dynamic parameters (absolute position, acceleration/deceleration, opening/closing speed, enable status), upstream and downstream choke pressures, and the position and command signals of the electric slab-gate valves in the diverter manifold. Choke-motion data are communicated in real time via Modbus TCP/IP between the PK controller and the servo-motor drive. Upstream/downstream choke pressures are acquired through FOUNDATION Fieldbus (FF) by means of an FF-bus data card installed in the PK controller. Electric slab-gate valve positions and commands for the diverter manifold are likewise interfaced to the PK controller via FF.

Micro-flow-module electro-hydraulic system: This system collects and processes data from the Coriolis mass flow meter and the dual-frequency Doppler non-intrusive flow meter. The mass flow meter transmits flow rate, density, and temperature via RS-232 serial communication. The dual-frequency Doppler meter outputs measurement data as a 4–20 mA analog signal. Consequently, the micro-flow electro-hydraulic system incorporates a serial-port data-receiver card and an analog-input (AI) data-acquisition card. An additional FF-bus data-acquisition card is included to capture upstream and downstream pressures for auxiliary influx/loss detection.

Electro-hydraulic integration system

Both the automatic MPD and the auto-driller control systems use PLC-based controllers located, respectively, at the well site and inside the driller's cabin. To overcome the long-distance data-transmission challenge, a dedicated electro-optical converter module was developed. High-speed fiber-optic links connect the MPD controller(s) to the automated-rig controller(s), creating a dedicated physical data highway. All MPD sub-modules and each automated-rig control sub-module are interfaced to the electro-optical converter, laying the groundwork for top-level data acquisition and unified control.

Unified communication layer for integrated MPD and automated-rig control

The fusion-link communication layer is built around the DeltaV DCS software, which is interfaced to the electro-optical converter. A host workstation is configured as the master station; via Modbus TCP/IP, both the automatic MPD system and the automated-rig control subsystems are connected to this workstation. OPC UA read/write operations are employed to collect and transmit data from every module, thereby solving the integration-communication challenge between the automated rig and the automatic MPD system.

Multi-source data fusion and exchange interface

On the basis of the three-layer integrated communication system, a multi-source-data fusion and exchange interface must be built between the automatic managed-pressure drilling (MPD) system and the automated drilling rig. Control signals, measured data, and feedback data from the rig PLC, auto-driller PLC, MPD DCS, and other sources are transmitted through WITS, RS-232 serial, Modbus TCP/IP, and OPC UA. Therefore, a unified interface that fuses these heterogeneous protocols has been designed and developed.

WITS Data Transmission

The Wellsite Information Transfer Specification (WITS) is a communication format composed of discrete data records. Each record is independent of the others, has its own trigger variable, and is sampled at its own interval. WITS also defines a mechanism for a remote computer to send commands to the transmitter to set or change parameters, including data types and transmission intervals.

In addition to the transmission format, WITS defines a basic set of data types while allowing user-defined records.

Serial Communication Basics

Serial communication transmits and receives bytes bit by bit. Although slower than byte-wide parallel communication, it allows data to be sent on one line while simultaneously receiving on another. The critical parameters are baud rate, data bits, stop bits, and parity.

Modbus TCP/IP Communication

Modbus is a universal industrial protocol that enables controllers to communicate with each other or with other devices over networks such as Ethernet. The protocol uses master-slave technology: the master initiates queries and operations on the slaves. The physical interface can be serial (RS-232/RS-485) or Ethernet. In the current MPD system, all modules communicate via Modbus TCP/IP over Ethernet.

Real-Time Data Interface Program

The real-time interface integrates TCP/IP, serial, Modbus RTU, and OPC data streams, enabling free data flow among the rig, auto-driller, MPD modules, mud-logging, and MWD/LWD systems. This resolves the data-interaction challenge. The functional design of the integrated real-time interface is shown in Fig. 2.

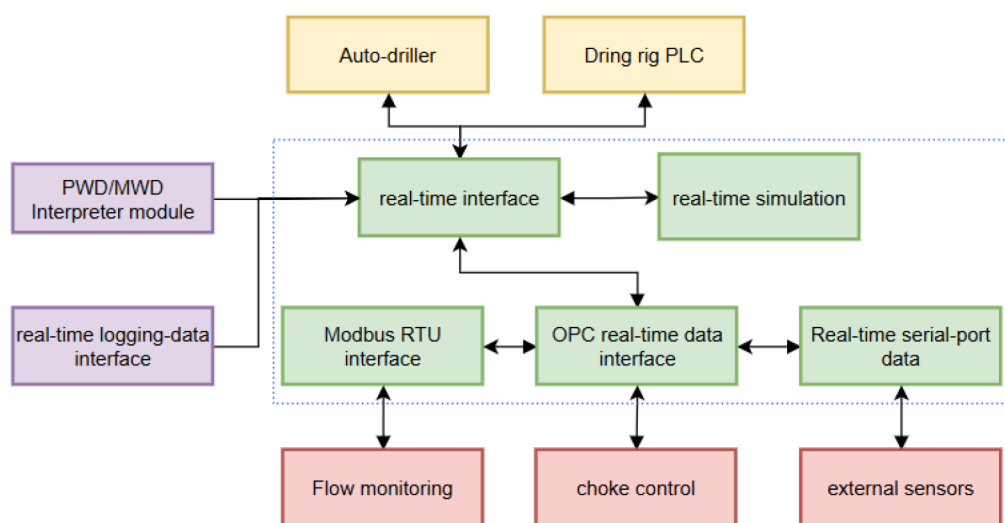


Figure 2—Multi-source real-time data exchange interface data flow

A high-pressure, high-precision servo-electric choke valve

By adopting a servo-motor design, an integrated electric actuator was developed to enhance choke-valve positioning accuracy, while a refined valve-profile geometry was optimized to boost pressure-bearing capacity, providing a core component for precise pressure regulation under complex operating conditions.

A custom-engineered hollow, compact servo motor was innovated with a co-axial "reducer - brake - planetary roller screw" layout that directly reduces axial length by over 40 %. A large-aperture harmonic drive delivers high torque density (700 N·m) and, in tandem with the planetary roller screw, converts rotary motion into linear motion, enabling the actuator to deliver a continuous thrust of 21 t—serving as the key element for fine pressure control, as shown in Fig. 3.



Figure 3—The schematic diagram of the integrated servo actuator

To meet ultra-high-pressure precision requirements, the valve geometry was optimized through numerical simulation based on the "throttling-pressure vs. opening linearity" model. A high-strength nickel-based hard alloy was selected, and—leveraging the integrated electric actuator—a servo-electric choke valve rated for 105 MPa (operating range 0–35 MPa) was created, satisfying the high-pressure demands of ultra-deep, complex formations, as shown in Fig. 4.

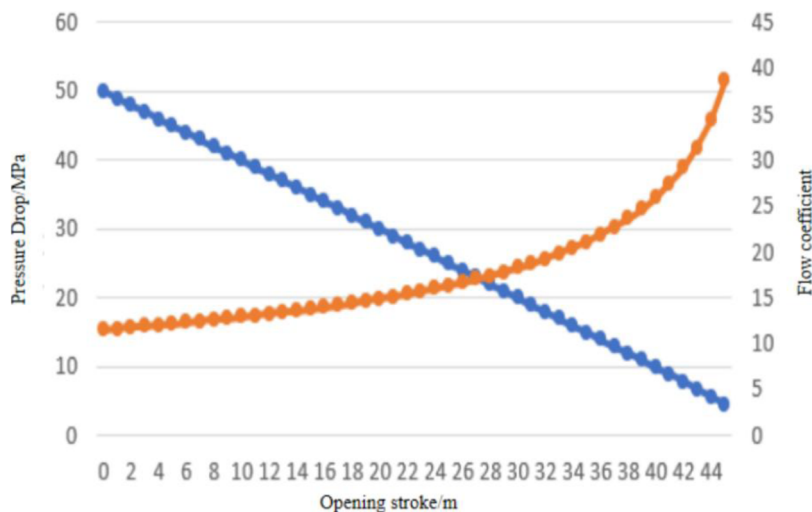


Figure 4—Linear pressure-drop curve of the working choke valve

Rectifier–inverter current control regulates choke motion speed. A seven-segment S-curve and T-curve acceleration algorithm was developed. Using the servo-motor principle, a multi-turn absolute encoder was chosen, achieving an angular resolution of 19.8 arc-seconds. Combined with the planetary roller screw, this yields a linear resolution of 0.0006 mm, realizing precise choke positioning with a positional accuracy of ± 0.01 mm, as shown in Fig. 5.

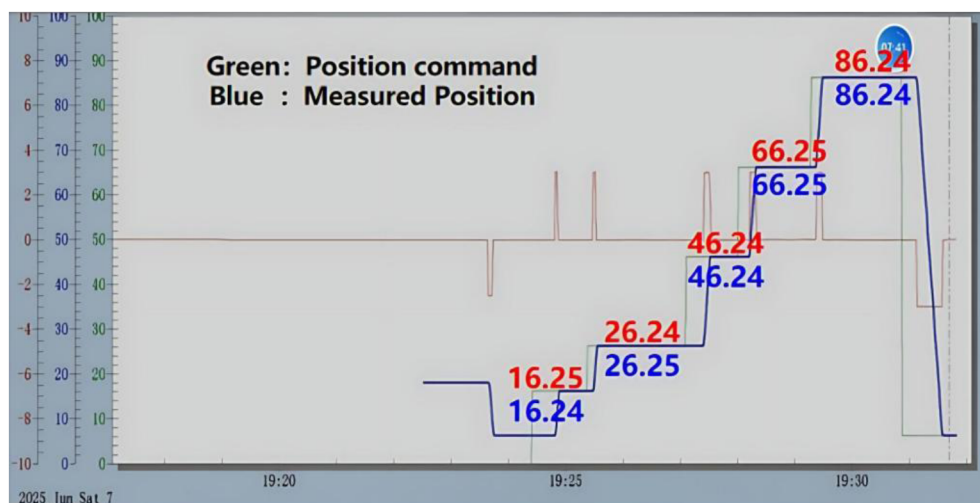


Figure 5—Choke valve precision positioning test

Multi-equipment-linked intelligent pressure regulation system

The system comprises a choke-valve feedback control module, a gate-valve on/off control module, a diverter-manifold sequential control module, a micro-flow monitoring module, an operating-mode configuration module, and an initialization-parameter module.

Choke-valve feedback control module

In DeltaV Control Studio, measurement blocks for choke position and upstream/downstream choke pressures are created, along with an analog block that correlates choke position with pressure change. A PID control block is then added: the measurement blocks feed the PID as inputs, and the PID output adjusts the choke position to regulate pressure. The innermost loop is position control—implemented with a PID block for closed-loop position feedback—while the outer loops are wellhead-pressure and bottom-hole-pressure control, each also implemented as a PID block. The output of the outer PID is cascaded to the set-point of the inner position PID, achieving high-precision dual-cascade control.

Gate-valve on/off control module

Within DeltaV Control Studio, a gate-valve control module is created. By reading the current open/closed status of the gate valve and comparing it with the commanded state (open or close), the module generates the corresponding digital I/O command to drive the valve to the desired position.

Diverter-manifold sequential-control module

On the basis of the choke-position and gate-valve control modules, a sequential-control program for the diverter manifold was developed. During managed-pressure operations, the program executes the following steps in order: Open the diverter-manifold bypass line to redirect standpipe flow to the automatic choke inlet. Dynamically adjust the choke opening to compensate for annular pressure losses caused by the diversion. After all flow has been transferred to the choke system, close the gate valve on the standpipe line. Once the gate valve is confirmed closed, open the standpipe relief valve to bleed off residual pressure between the standpipe and the top drive.

Micro-flow monitoring module

Based on real-time inlet and outlet flow rates, instantaneous-flow-change and cumulative-volume-change blocks are created, thereby realizing a micro-flow monitoring module for detecting minute variations in flow imbalance.

Based on the above work, a system-monitoring main screen was created in DeltaV Operate's Configure mode. The main interface integrates real-time data monitoring and control capabilities for the automatic MPD module, the automated drilling rig, and all auxiliary equipment; the resulting HMI is shown in Fig. 6.

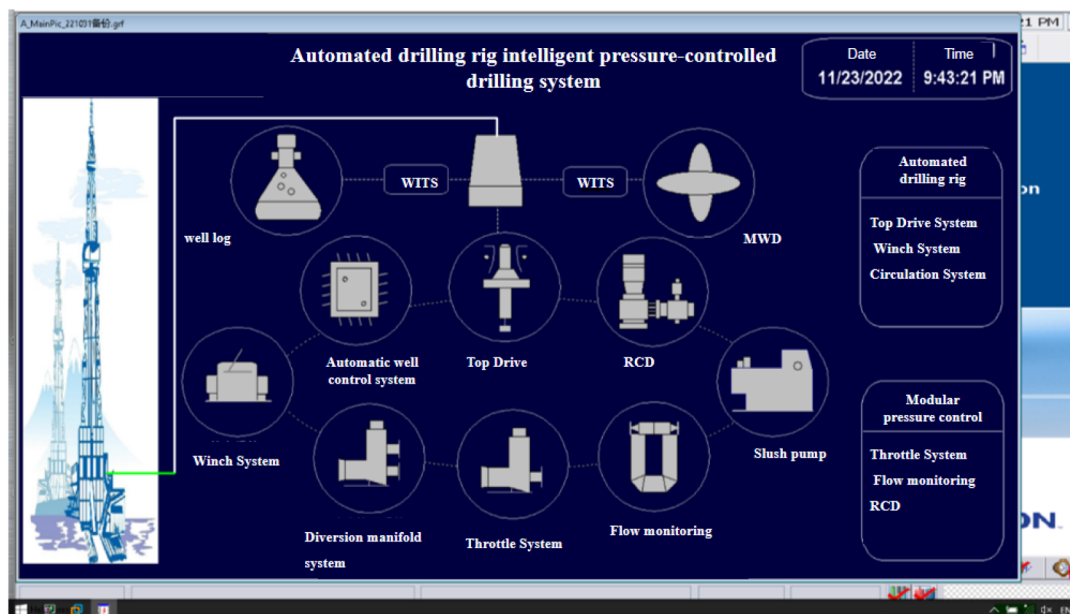


Figure 6—Interface of the Multi-Equipment-Linked Intelligent Pressure Regulation System

Application and analysis

The integrated well-control system has been deployed on numerous wells in the SL Block of SINOPEC. During operation, it has demonstrated real-time kick and loss detection with immediate alarms and has been successfully integrated with 7000 m automated drilling rigs. To validate the effectiveness of the proposed system, the field applications of Well NY and Well JY are presented in detail.

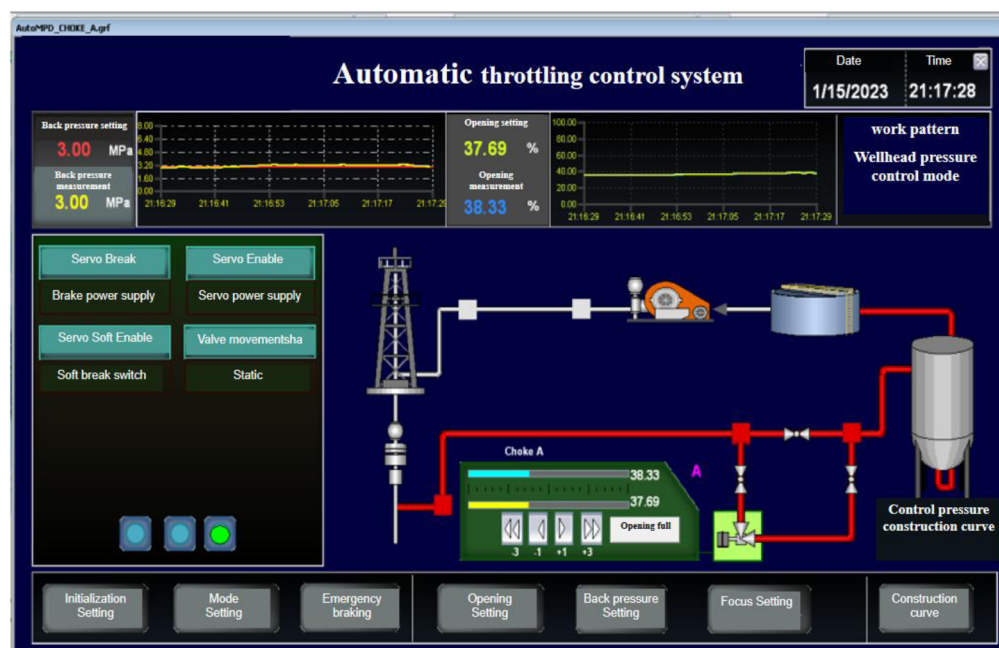


Figure 7—Managed Pressure Drilling Monitoring Interface

NY-Well was drilled with a 7000 m automated rig developed by the Machinery Company. Fig. 8 shows the MPD monitoring interface, and Fig. 8 illustrates on-site surveillance via the MRO IoT system. Throughout the drilling phase, the applied back-pressure was maintained between 0.5~5.0 MPa, keeping the bottom-hole ECD within 1.95~1.96 g/cm³ and sustaining a near-balanced drilling condition. This successfully resolved the challenges of pressure-sensitive reservoirs, narrow mud-weight windows, and simultaneous loss/influx scenarios in the block, achieving seamless integration with the automated rig.

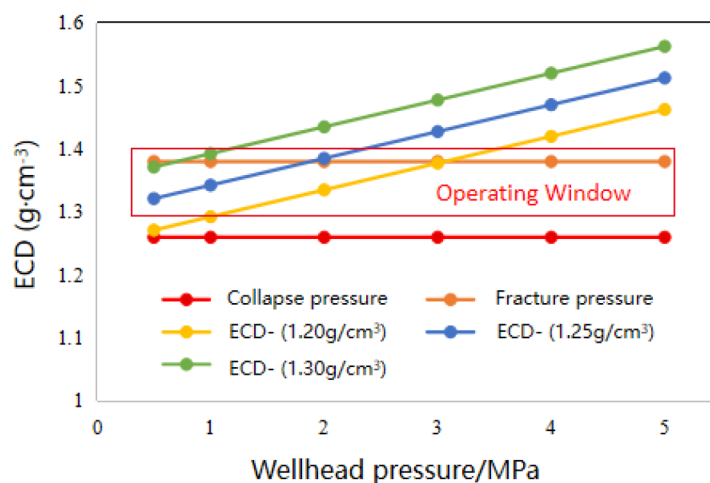


Figure 8—JY-well MPD Coring Operating Window Diagram

After stimulation, the coring well cluster is densely intersected by natural fractures and induced fracture networks. Prolonged production has weakened formation strength and raised collapse pressure, resulting in high risks of lost circulation, wellbore instability, and kicks. Managed pressure drilling (MPD) was applied in JY-Well to reduce mud weight by 0.25~0.35 g/cm³ and to maintain surface back-pressure between 0.5~2.5 MPa. Bottom-hole ECD was precisely controlled at 1.23~1.25 g/cm³ and 1.30~1.36 g/cm³ in separate intervals. Despite peak total-gas readings up to 10 %, no losses, kicks, or collapses occurred throughout the operation, as shown in Fig.8.

Conclusions

The newly developed intelligent MPD system has been successfully deployed in more than 20 shale oil and gas wells of SINOPEC. Field results demonstrate that it delivers precise pressure control with a full-process error of less than 30 psi. In addition, the accompanying software platform enables seamless data exchange among the rig circulation system, auto-driller, MPD modules, and real-time mud logging, eliminating previous data-interoperability barriers. An integrated pressure-calculation and remote-control system covering drilling, cementing, and well-kill operations was further developed, providing full-condition wellbore pressure monitoring and regulation. This has realized intelligent, coordinated tuning between MPD equipment and the automated drilling rig, and field application has proven its ability to achieve accurate pressure control through multi-device synergy on the rig floor, thereby advancing the large-scale deployment of fine MPD technology.

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